Nocturia & Male Incontinence

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 112	Nocturia
Ch 123	Surgical Procedures for Sphincteric Incontinence in the Male
★	Putting it together

112 Nocturia

1 · Definition and terminology

- **Nocturia** — International Continence Society (ICS) 2019: number of voids during the **main sleep period** (intending to sleep → intending to rise), set by the **individual's own schedule** (shift workers included), not the clock.
- Does **not exclude 1 void per night**; does **not incorporate bother**.
- ICS 2019 dropped "wake **to void**" (2002): "non-urological" causes are **leverage for interdisciplinary care, not grounds to exclude**.
- Guidance: European Association of Urology (EAU) and ICS; American Urological Association (AUA) has none — its overactive bladder (OAB) guideline calls **isolated nocturia a separate entity**.

Key diary terms

Term	Definition / formula	Meaning
Actual nightly voids (ANV)	Voids in main sleep period	Severity count
First uninterrupted sleep period (FUSP)	Sleep onset → first night void (h)	Key sleep measure
Nocturnal urine volume (NUV)	Night voids **+ first morning void**	Volume made during sleep
Nocturnal polyuria index (NPi)	NUV / 24-h volume	Proportion made at night
Nocturnal urine production (NUP)	NUV / hours of sleep	Hourly rate
Maximum voided volume (MVV); nocturnal MVV (NMVV)	Largest void in 24 h; in sleep period	Proxies for functional capacity
Nocturia index (Ni)	NUV / MVV	>1: night output exceeds capacity
Modified Ni (MNi)	NUV / NMVV	Night data only
Predicted nightly voids (PNV)	Ni − 1	—
Nocturnal bladder capacity index (NBCi)	ANV − PNV	>0: voids below capacity

Trap: first morning void is **split** — volume goes into NUV, but it counts as daytime frequency, not nocturia.

2 · Why nocturia matters

- **Common**: up to **3 in 5** people >70 years void ≥2 times nightly; age is the most robust correlate. Boston Area Community Health survey MCQ: higher in Black and Hispanic people, partly via socio-economic status.
- **Bother**: most with ≥**2** voids; but **any nocturia that brings a patient to a urologist is clinically significant**.
- **Sleep**: **commonest cause of adult sleep disruption**. Restorative slow-wave sleep (N3) clusters early; most first voids occur within **2–3 h** of sleep onset and break it → daytime fatigue.
- **Falls and hip fracture**: among the LUTS most strongly linked.

- **Systemic disease**: may herald cardiometabolic (strongest), renal, endocrine or neurological disease; ≥2 voids/night → higher mortality (causality unproven).

> **KEY POINT**
> The first void breaks restorative slow-wave sleep; the symptom may be the visible edge of systemic disease.

3 · Pathophysiology — production, capacity and sleep

Multifactorial — **"Sleep CALM"**: **S**leep disorders, **C**omorbidities, **A**ctions (behaviour, lifestyle), **L**ower urinary tract dysfunction (LUTD), **M**edications. Idiopathic nocturia is small but real.

- Every factor upsets **night production**, **functional capacity** or **sleep**; at capacity: **wake to void or enuresis**. Sleep disorders cause "convenience" voids (**bidirectional**).

Nocturnal polyuria (NP) and global polyuria

Criterion	Threshold	Comment
NPi (ICS)	> **20%** young to > **33%** older adults	Most used; poor discrimination
NUP (ICS alternative)	> **90 mL/h**	More specific for production-driven nocturia
Global polyuria	> **40 mL/kg/24 h** (>2.8 L at 70 kg)	Uncommon, serious — **always investigate**

- **Krimpen study** MCQ: NPi >33% in 92% of older men with ≥2 voids but also 70% **without** nocturia.
- Excess night production contributes to **most adult nocturia**; mechanisms often coexist.

Functional capacity and the mismatch

- Normal capacity usually >300–400 mL; **150–300 mL** suggests diminished capacity — **24-h** (all voids small) or **night-specific** (partly convenience voids).
- **Ni is the strongest diary determinant of severity and treatment response**. PNV = Ni − 1 because the last night volume leaves as the first morning void.

> **ALGORITHM**
> NUV 800 mL, MVV 200 mL → Ni = 4 → PNV = 3 → diary shows 6 night voids → NBCi = **3** (voiding well below capacity)

Investigate diminished night capacity when NBCi > **1.3**; > **2** strongly associated with severe nocturia.

4 · The bladder diary

EAU (strong): for female LUTS and male LUTS with storage symptoms or nocturia.

> **ALGORITHM**
> Measuring container → time and volume of every void → time, volume, type of every drink → mark bedtime (intending sleep) and rising → urgency grade and leakage per void

- ICI urgency codes: **0** no need (social), **1** normal desire, **2** urgency that passed, **3** urgency, dry, **4** urgency with leak.
- **Diary ANV is the gold standard for severity**; questionnaires overestimate. **Intent governs**: lying awake intending to sleep is still the sleep period.
- **FUSP is the most important sleep measure** — outcome measure and target.
- Duration: EAU **3 days** (Male: at least 3). "Diary fatigue": fewer than half complete.

- **Nocturnal-only diary**: MNi substitutes for Ni but **cannot diagnose global polyuria**; not recommended for women by EAU.

> **KEY POINT**
> Every cause works through production against capacity, gated by sleep; their ratio (Ni) predicts severity better than either alone.

5 · Evaluation — the Sleep CALM work-up

Basic work-up in all → tailored tests, multidisciplinary referral. **TANGO** questionnaire screens all domains, including falls.

Factor	History / examination	Tests (basic → ancillary)
Sleep	Habits, apnoea, restless legs, pain, anxiety, depression	STOP-BANG, sleep diary → polysomnography, sleep specialist
Comorbidities	Blood pressure, volume overload, neurological signs	Metabolic panel → HbA1c, ECG, echocardiography, natriuretic peptide, albumin:creatinine ratio, calcium, thyroid, water deprivation test
Actions	Fluid, alcohol, caffeine, sodium; BMI	Fluid intake diary
LUTD	Concurrent LUTS, prior surgery; abdominal, genital, rectal examination	Urinalysis, PVR → PSA, uroflowmetry, cystoscopy
Medications	Reconcile drugs and **dose timing**	—

- **Trap**: urology is one row of five. First-line = **optimise every Sleep CALM abnormality**.

6 · Sleep disorders

Obstructive sleep apnoea (OSA): best-described link; frequent nocturia predicts severe OSA.

Arm	OSA mechanism	Continuous positive airway pressure (CPAP)
Production (main)	Hypoxia → pulmonary vasoconstriction → ↑ right atrial transmural pressure → ↑ **atrial natriuretic peptide (ANP)** → natriuresis	↓ night production
Capacity	Intermittent hypoxia → oxidative stress → detrusor instability	Less detrusor overactivity
Sleep	More light, less slow-wave sleep	↑ slow-wave sleep

- CPAP reduces nocturia by up to ≈**2 voids** in moderate–severe OSA — among the **largest effects of any intervention**. EAU Female (strong): suspected OSA → sleep clinic.
- **STOP-BANG**: Snoring, Tired, Observed apnoea, Pressure, BMI >**35**, Age >**50**, Neck >**40 cm**, Gender male; ≥**3** = high risk; highly sensitive.
- Ask the **primary motive for getting up** (urge vs already awake); screen OSA, restless legs, parasomnias.
- **Melatonin** may suit older adults; behavioural insomnia treatment reduced night voids.

> **KEY POINT**
> OSA raises production, lowers capacity and fragments sleep — hence one of the largest nocturia responses to treatment.

7 · Comorbidities

- Act mainly via **excess night production**; nocturia may be the **first sign** of hypertension, metabolic syndrome, coronary disease, diabetic end-organ damage or early kidney disease. EAU: consider systemic causes in all; even **non-bothersome nocturia merits a frequency–volume chart**.
- **Volume overload**: leg fluid mobilised on lying down (heart failure).
- **Hypertension**: blood pressure normally dips **10–20%** in sleep; **non-dipping** (renin–angiotensin–aldosterone dysregulation) → **nocturnal natriuresis**. Hydrochlorothiazide helped men with α-blocker-refractory LUTS.

Disorders of water homeostasis

Arginine vasopressin (AVP) acts on collecting-duct **V2 receptors**; defects give **24-h polyuria**.

Condition	Defect	Causes
Central diabetes insipidus (DI)	↓ AVP secretion	Brain tumour, haemorrhage, trauma, autoimmune
Nephrogenic DI	Renal AVP insensitivity	Drugs, electrolytes, renal failure
Primary polydipsia	AVP suppressed by intake	Idiopathic; psychiatric

Arginine-stimulated **copeptin** appears superior to the water deprivation test. Evaluate every global polyuria (neoplastic, traumatic, inflammatory, psychiatric causes).

> **KEY POINT**
> Comorbidity acts through the kidney (third-spaced fluid, nocturnal natriuresis, failed antidiuresis). Global polyuria and non-bothersome nocturia both warrant looking beyond urology.

8 · Actions — lifestyle and behaviour

- **Timing matters more than amount**: water → diuresis within ≈1 h; caffeine, alcohol within 2–3 h (both disturb sleep); evening protein → hyperfiltration.
- Restrict fluid, caffeine, alcohol from **3–4 h** before sleep; older adults drink more by day. High baseline intake gains most.
- Also: **sodium** restriction, weight loss, exercise, **compression stockings**, bladder training, pelvic floor muscle training (PFMT), treat constipation.

9 · Lower urinary tract dysfunction

- Nocturia accompanies bladder outlet obstruction (BOO), OAB (up to **85%** of trial participants), detrusor underactivity, neurogenic bladder and genitourinary syndrome of menopause (GSM) → *Voiding Dysfunction book, Ch 110*.
- **Isolated nocturia is not a LUTS** — not expected to respond to bladder or prostate drugs.
- Urinalysis in all; **PVR in both sexes**; PSA only if it changes management. †Urodynamics not routine → *Voiding Dysfunction book, Ch 107*.

Benign prostatic obstruction

- Nocturia is a secondary endpoint of benign prostatic hyperplasia (BPH) trials: **α-blockers** carry most effect; 5α-reductase inhibitors (5ARIs) alone weakest. Gains are fractions of a void with large placebo response — **statistical, not clinical, significance** *→ Voiding Dysfunction book, Ch 113*.
- **Surgery**: TURP beat microwave, α-blocker, watchful waiting; newer techniques not proven better. **Counsel** that nocturia may respond sub-optimally.

Overactive bladder

- Night symptoms most refractory. Tolterodine (night voids primary endpoint) negative overall but reduced urge-related voids; **fesoterodine** (nocturnal urgency primary endpoint) positive. **Raised NBCi** → best response to evening dose.
- β3-agonists: marginal (≈0.1–0.2 voids vs placebo).
- **Behavioural therapy** first; matches or beats drugs.
- Antimuscarinics are on the **Beers Criteria** for ≥65 years — the core nocturia population.

Therapy	EAU Male LUTS	EAU Female LUTS
α1-blocker	Offer for nocturia with LUTS (weak)	—
5ARI	Offer with LUTS and prostate > **40 mL** (weak)	—
PDE5 inhibitor	**Do not offer**	—
Antimuscarinic	Offer with OAB (weak)	Offer after counselling (**strong**)
PFMT; vaginal oestrogen	—	Offer (**strong**); offer (weak)

KEY POINT
LUT drugs act on capacity and outflow, but most nocturia is production-driven — effects are statistically significant, clinically small. Treat the mechanism, not the organ.

10 · Medications causing nocturia

Mechanism	Drugs
↑ Urine output	Diuretics; SSRIs (block antidiuretic hormone); **calcium channel blockers** (oedema, ↑ ANP); tetracycline; lithium (↓ aquaporin-2); **SGLT-2 inhibitors** (glycosuria)
Insomnia / CNS	Stimulants; α-/β-blockers; salbutamol, theophylline; corticosteroids, thyroxine; SSRIs; phenytoin
Bladder toxicity	Ketamine; tiaprofenic acid; cyclophosphamide

α-blockers both treat nocturia and may cause insomnia. Adjust timing first, or substitute.

11 · Management algorithm and refractory nocturia

ALGORITHM
History + examination + diary + basic tests → identify Sleep CALM factors → optimise them (multidisciplinary) → conservative measures → antidiuretic or timed diuretic in selected refractory patients — diary at every step

- **Multicomponent therapy**: treat interacting causes together (EAU **"Shared Care Pathway"**).

- **Target night volume even when capacity is small**: small-capacity patients improved as NUV fell with capacity unchanged.

Desmopressin

- P3 Desmopressin (DDAVP) — *selective V2 agonist, no pressor effect; hyponatraemia risk factors (age ≥65, Na <135, CrCl <60, high dose), most in first 2–3 weeks; EAU <65 / low dose >65; contraindications*
- A **blunted nocturnal AVP rise** occurs in some NP patients. Desmopressin is the **only US-approved nocturia drug**: **≥2 voids per night due to NP** (level 1b).

	Nasal spray	Orally disintegrating tablet (ODT)
Dose	1.66 mcg if <65, no risk; ≥65 or at risk → **0.83 mcg**	**Sex-specific**: women **25 mcg**; men **50 mcg**
Timing	≈30 min before bed	Sublingual ≈60 min before bed

- **Women more sensitive**: V2 receptor gene on the **X chromosome**. In men 75 mcg added hyponatraemia, not efficacy.
- †Minimal fluid from 1 h before to 8 h after the dose. †Also contraindicated with systemic/inhaled glucocorticoids and eGFR <50 mL/min.
- Hyponatraemia defined as sodium <**130 mmol/L**; most within **3 weeks**.
- **Sodium monitoring**: EAU — baseline → **day 3** → day 7 → 1 month → periodically; US label — baseline → within 7 days → ≈30 days. EAU Female: avoid if baseline sodium low. Proposed: baseline sodium in all, stop if <135, repeat at 7 and 30 days only if ≥65/at risk.
- **Low dose** still works and avoids antidiuresis outlasting sleep. **Age 65 is the pivot.**

Timed diuretics

- Furosemide acts 6–8 h: evening dosing **causes** NP; **morning or early-afternoon** dosing treats it (level 1b). Small trials (bumetanide 4–6 h, furosemide 40 mg 6 h before bed) consistently cut voids. Monitor sodium.
- **Trap**: afternoon furosemide + bedtime desmopressin worked in a trial, but loop diuretics **contraindicate** desmopressin in US labelling.

> **KEY POINT**
> Refractory nocturia: move urine production out of the night. The price is sodium balance — dose, age and monitoring define safe use.

12 · Landmark trials

Trial	Design	Result	Take-home
MTOPS	Doxazosin, finasteride, combination, placebo	Doxazosin and combination beat placebo; finasteride did not	α-blocker drives benefit
VA Cooperative	Terazosin, finasteride, combination, placebo	Only terazosin separated	5ARI alone does not help
CombAT	Tamsulosin, dutasteride, combination	Combination −0.5 vs −0.3/−0.4	Significant but small
Fesoterodine RCT	Nocturnal urgency primary endpoint	Positive	Helps urgency-driven night voids
Nasal spray phase III	0.83/1.66 mcg vs placebo	Both beat placebo	Basis of approval

Trial	Design	Result	Take-home
ODT sex-specific RCTs	Women 25; men 50/75 mcg	Positive; 75 mcg more hyponatraemia	Sex-specific dosing

123 Surgical Procedures for Sphincteric Incontinence in the Male: Artificial Urinary Sphincter and Perineal Slings

1 · Pathophysiology and device mechanisms

- Male **sphincteric (stress) urinary incontinence (SUI)** needs failure of **both** sphincters. Radical prostatectomy (RP), the commonest cause, removes the internal sphincter, leaving the rhabdosphincter. Others: radiotherapy (RT), pelvic fracture urethral injury, neuropathy; treating vesicourethral anastomotic stenosis (VUAS) can damage the membranous urethra.
- Work-up and sling-vs-sphincter overview → *Voiding Dysfunction book, Ch 106*.

Device	Mechanism	Site	Adjustable	Needs residual sphincter
Artificial urinary sphincter (AUS, AMS 800)	Circumferential compression, patient-cycled	Bulbar urethra or bladder neck	No	No
Advance / Advance XP	**Repositioning**, lengthening; no compression	Bulbar, transobturator	No	**Yes**
Virtue (quadratic)	Prepubic compression + elevation	Bulbar, four arms	No	Less dependent
ATOMS (adjustable transobturator male system)	Cushion compression	Bulbar, transobturator	Yes — scrotal port	Less dependent
ProACT	Two para-urethral balloons	Bladder neck	Yes — scrotal ports	—

- **AUS**: cuff (≈2 cm long), scrotal pump, pressure-regulating balloon (PRB) setting cuff pressure. Cuff refills passively †over minutes. Men without detrusor contraction can void by straining/Credé.
- Transobturator sling raises leak point pressure **without obstruction** by elevating the bulb and rhabdosphincter.

> **KEY POINT**
> Every device compresses the urethra or repositions a functioning sphincter. Trap: ATOMS and Virtue are "slings" but compress — only the Advance depends absolutely on a coapting residual sphincter.

2 · Indications and device selection

- Indication: **irreversible sphincter incompetence**, bothersome incontinence (UI) after conservative care. Do not delay beyond **6 months** for severe SUI, especially with a cystoscopic sphincter defect.
- 24-h pad weight: mild **<150 g**, moderate **<400 g**. AUS preferred if >250 g or after RT (MASTER → *Ch 106*).

Stratum (AUA/SUFU)	Recommendation
Severe UI or **any RT**	AUS first-line; sling only with counselling
Moderate, no RT	AUS or male sling; adjustable balloons in specialised centres
Mild, no RT	Continue PFMT, sling, possibly AUS

- Counsel: ~1 thin pad/day; AUS lifespan **7–10 years**; failed slings mostly go to AUS.
- **Sling needs**: mild–moderate SUI, contractile bladder (hypocontractile → AUS), non-irradiated urethra, coapting sphincter.
- **AUS needs**: dexterity and cognition, healthy urethra, adequate compliance.

Absolute — need an alternative	Relative
Low compliance threatening kidneys; low capacity (urgency incontinence); urethra too poor → clamp, catheter, bladder neck closure, diversion	Frequent transurethral work — bladder cancer, refractory VUAS, radiation cystitis (erosion, infection)

- Radiation cystitis: hyperbaric oxygen and **6 months** bleeding-free before AUS.

> **KEY POINT**
> Each device's contraindication mirrors the other's requirement.

3 · The high-risk ("fragile") urethra

- Strong erosion factors: **prior RT, prior erosion/infection, urethroplasty**; also hypogonadism, catheterisation, 3.5-cm cuff after RT, tandem distal cuff. Debated: coronary disease, age >80, diabetes, smoking.
- Prior erosion: 5-year repeat erosion approaches **50%**. Urethroplasty worse than RT — transection divides bulbourethral arteries.
- Countermeasures: optimise medically, **transcorporal (TC) cuff**, PRB **51–60 cm H_2O**, cuff 0.5 cm larger, nightly deactivation.

> **KEY POINT**
> Erosion = spongiosal ischaemia under constant pressure. Countermeasures add tissue (TC tunica) or lower pressure — never squeeze harder.

4 · Operative technique

- Low lithotomy, chlorhexidine–alcohol; sterile urine. IV antibiotics within **60 min** of incision: aminoglycoside 5 mg/kg + first/second-generation cephalosporin or vancomycin.
- Bulbar cuff site: **proximal bulbar urethra, proximal to corporal fusion** — largest spongiosum, cuff not opened by sitting; distal sites kept for revision.

> **ALGORITHM**
> Midline perineal incision → bulbospongiosus divided → incise Buck fascia at crural divergence → 2-cm tunnel round urethra → measure → cuff (usually 4.0–4.5 cm) → PRB preperitoneal (lower-quadrant or via external ring), filled **23 mL** → pump subdartos in anterior scrotum → cycle; urethroscopy → leave **deactivated 4–6 weeks**

- Sizing: consistent method matters; less tight cuffs continent better. **3.5-cm cuff** raises erosion after RT. Avoid blind ring PRB route after mesh hernia repair or cystectomy.
- †Deactivate (open) the cuff before any later catheterisation or endoscopy.

PRB (cm H_2O)	Use
51–60	High-risk bulbar (RT, prior erosion, atrophy) — trades continence for safety
61–70	Standard bulbar cuff
71–80	Bladder neck cuff

- **Tandem cuff**: ≥1.5–2 cm distal, Y-connector, deactivate 6 weeks. Not first-line — distal cuff erodes more.
- **Transcorporal cuff** (Webster 2002 MCQ): dissects through corpora so ventral tunica stays on dorsal urethra, adding bulk. For urethra <3.5 cm, revisions, prior erosion/urethroplasty. Erections usually preserved; more transient retention; erosion benefit unproven.
- **Bladder neck AUS**: prostate intact (neuropathic, myelomeningocele); **contraindicated after RP**; cuff ≥8 cm, PRB 71–80; less erosion.

ALGORITHM

Advance XP: perineal incision → open bulbospongiosus → divide perineal body (releases bulb) → helical trocar outside-in below adductor longus tendon → mesh fixed to spongiosum → tension under cystoscopy until mucosa coapts (bulb moves 3–4 cm proximally)

- **ATOMS**: bulbospongiosus **left intact**; scrotal port; adjusted in clinic. **ProACT**: fluoroscopic placement at bladder neck.

5 · AUS complications and management

ALGORITHM

Retention: confirm cuff deactivated → small catheter 24–48 h → fails second trial → **suprapubic cystostomy** (avoid PRB) → weeks of retention = undersized cuff → replace cuff

- **Infection 1–3%** (up to 10% high-risk); staphylococci. **Scrotal pain = key early sign**. Antibiotics fail → explant all; reimplant 3–6 months. Immediate salvage (remove, irrigate, new device) only if not eroded, septic or immunosuppressed. Coated devices have not reduced infection.
- **Erosion 2–15%** (up to 35% with several risk factors): remove **all components**. Urethral loss <50% → catheter ~3 weeks; >50% → in situ ventral approximation or anastomosis. Post-erosion stricture 8–35%. Reimplant at **3–6 months** at a **new site**, TC ± low-pressure PRB.
- **Atrophy**: **gradual** recurrent leak, common revision cause. Downsize, relocate (TC distally) or tandem. **Never simply raise PRB pressure**.
- **Mechanical failure**: <3 years replace failed part; **>3 years replace entire device**.

Recurrent leak after AUS	Clue → confirm
Deactivated	Sudden; reactivate
Fluid leak	Sudden; PRB shrinks on imaging
Atrophy	Gradual; PRB volume normal (>20 mL)
Erosion / detrusor overactivity	Cystoscopy / urodynamics

KEY POINT

Sudden = device, gradual = urethra. Manage every complication to keep a salvageable urethra for the next cuff.

6 · Sling complications and failure

Device	Commonest	Erosion / explant
Transobturator	Transient groin pain; retention/urgency 4–10%	0–0.4%
ATOMS	More complications	Erosion/infection ~10%; explant 10–16%
ProACT	**Perforation** (~5%); migration; revision 22%	~4%

- Cystoscopy in every worsening sling. Failed sling → **AUS = gold standard**, results as primary; repeat Advance only if AUS refused (~56%).

7 · Special circumstances

- Stricture at the cuff may herald erosion; VUAS above cuff → balloon or laser via flexible scope.
- AUS + penile prosthesis: synchronous = metachronous; metachronous → AUS first. **Climacturia**: sling or AUS; with ED → **mini-jupette** graft.

8 · Outcomes

- P2 P3 Post-prostatectomy incontinence; SUI: evaluation in females, role of UDS, and surgical therapy in males — *AUS social continence 55–77%, satisfaction ~90%, revision 10–29% at 3 years; Advance cure ~60%; ATOMS ~two-thirds dry*
- ProACT ~60–67% dry. Cure clusters at 55–65%; devices differ in durability, erosion and eligibility, not cure.

★ Putting it together

A · Nocturia: three diary questions

ALGORITHM

1. Global polyuria? >40 mL/kg/24 h → diabetes, insipidus, polydipsia; always investigate.

2. Nocturnal polyuria? Night fraction >20% (young), >33% (older) → heart failure, sleep apnoea, oedema, evening fluids; desmopressin with sodium checks.

3. Reduced night capacity? NBCi >1.3 → overactive bladder, obstruction, residual; treat it. Screen sleep in all ("Sleep CALM") (→ *Ch 112*).

B · Post-prostatectomy stress incontinence

KEY POINT

Pelvic floor training first; severe leakage → surgery by ~6 months. 24-h pad weight: mild <150 g, moderate <400 g. Mild–moderate, contractile detrusor, no radiotherapy → sling; >250 g, radiotherapy or failed sling → artificial sphincter. Erosion = spongiosal ischaemia (radiotherapy, prior erosion, urethroplasty): add tissue (transcorporal cuff) or lower pressure, not tighter. † Remove eroded/infected devices (→ *Ch 123; Ch 106*).